American Computer Science League

2022-2023 ● Contest 4: Short Problem Solutions ● Elementary Division

1. Graph Theory There are 7 current edges and a complete graph with 5 vertices has 10 edges. There are 3 missing ones which are AB, CE, and AD.	C. 3
2. Graph Theory There are 3 possible first stops, 2 second stops, and 1 third stop or 3*2*1=6 permutations. By inspection, they are HSM, HMS, SHM, SMH, MSH, and MHS.	B. 6
3. Graph Theory There are 4 ways to get a message from Amy to Evan. By inspection, using the first letter of each person's name, they are: ABDE, ACDE, ACBE, and ADBE.	D. 4
4. Graph Theory Cycles of length 3 in undirected graphs have 3 different vertices, start and end at the same vertex, and always go both ways. By inspection, they are JKMJ, KJLK, KMNK, and KNLK so there are a total of 4*2=8.	A. 8
5. Graph Theory For it to be traversable no matter where you start, all vertices must have an even degree. J, H, and F already do. Adding an edge that connects both G and K can't be done because one already exists. Therefore, it can't be done.	5. E